5.1 Generate the coefficients of an FIR High-Pass filter using Rectangular windowing techniques.

CODE:

```
clc;
clear;
close all;

%% Input Parameters
fp = 1000;        % High-pass cutoff frequency (Hz)
Fs = 8000;        % Sampling frequency (Hz)
N = 50;           % Filter order

% Normalized cutoff frequency
Wn = fp/(Fs/2);

%% FIR High-Pass Filter using Rectangular Window
b_rect = fir1(N,Wn,'high',rectwin(N+1));

disp('Rectangular Window Coefficients');
disp(b_rect');

%% FIR High-Pass Filter using Hamming Window
b_hamm = fir1(N,Wn,'high',hamming(N+1));

disp('Hamming Window Coefficients');
disp(b_hamm');
```

```
%% FIR High-Pass Filter using Hanning (Hann) Window
b_hann = fir1(N,Wn,'high',hann(N+1));     % Use hanning(N+1) if hann() is
unavailable

disp('Hanning Window Coefficients');
disp(b_hann');

%% Frequency Response

figure;
freqz(b_rect,1,1024,Fs);
title('FIR High-Pass Filter using Rectangular Window');

figure;
freqz(b_hamm,1,1024,Fs);
title('FIR High-Pass Filter using Hamming Window');

figure;
freqz(b_hann,1,1024,Fs);
title('FIR High-Pass Filter using Hanning Window');

%% Magnitude Response Comparison

[H1,f] = freqz(b_rect,1,1024,Fs);
[H2,~] = freqz(b_hamm,1,1024,Fs);
[H3,~] = freqz(b_hann,1,1024,Fs);
```

```
figure;
plot(f,20*log10(abs(H1)),'LineWidth',1.5);
hold on;
plot(f,20*log10(abs(H2)),'LineWidth',1.5);
plot(f,20*log10(abs(H3)),'LineWidth',1.5);
grid on;
xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');
title('Comparison of FIR High-Pass Filters');
legend('Rectangular','Hamming','Hanning');
axis([0 Fs/2 -100 5]);

%% Impulse Responses

figure;
subplot(3,1,1);
impz(b_rect,1);
title('Impulse Response - Rectangular');

subplot(3,1,2);
impz(b_hamm,1);
title('Impulse Response - Hamming');

subplot(3,1,3);
impz(b_hann,1);
```

```
title('Impulse Response - Hanning');

%% Pole-Zero Plots

figure;
subplot(1,3,1);
zplane(b_rect,1);
title('Rectangular');

subplot(1,3,2);
zplane(b_hamm,1);
title('Hamming');

subplot(1,3,3);
zplane(b_hann,1);
title('Hanning');

%% Display Information

fprintf('\nFilter Order = %d\n',N);
fprintf('Cutoff Frequency = %d Hz\n',fp);
fprintf('Sampling Frequency = %d Hz\n',Fs);
```

OUTPUT:

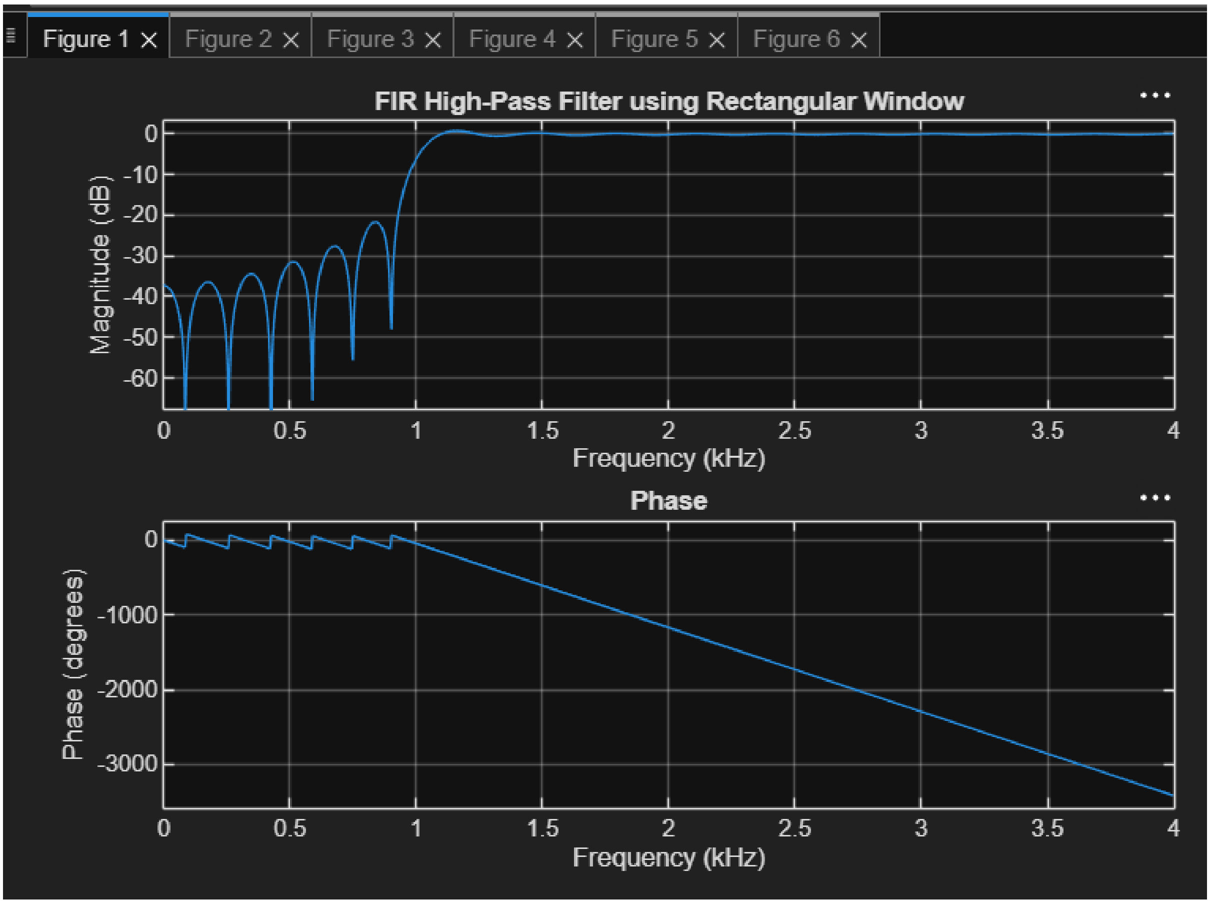


Figure 1
Figure 2
Figure 3
Figure 4
Figure 5
Figure 6
FIR High-Pass Filter using Rectangular Window
Magnitude (dB)
0
-10
-20
-30
-40
-50
-60
0
0.5
1
1.5
2
2.5
3
3.5
4
Frequency (kHz)
Phase
Phase (degrees)
0
-1000
-2000
-3000
Frequency (kHz)

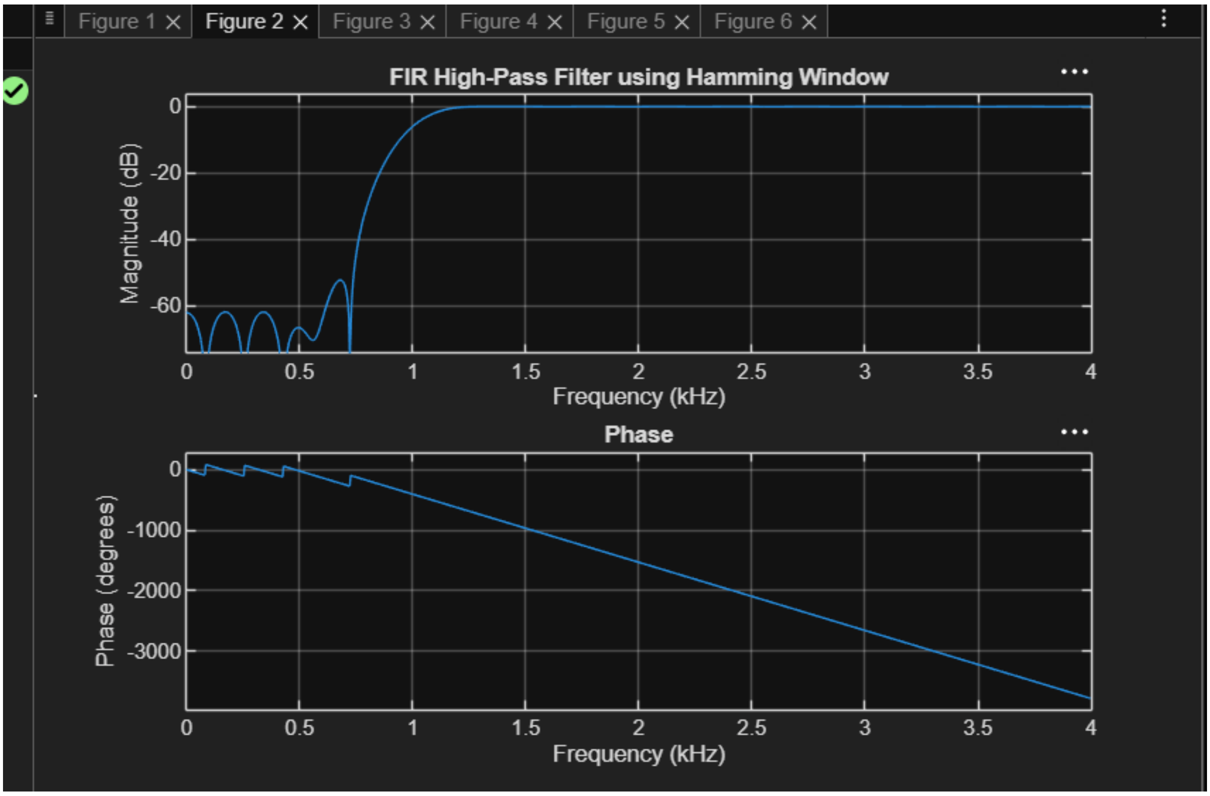


Figure 1
Figure 2
Figure 3
Figure 4
Figure 5
Figure 6
FIR High-Pass Filter using Hamming Window
Magnitude (dB)
0
-20
-40
-60
0
0.5
1
1.5
2
2.5
3
3.5
4
Frequency (kHz)
Phase
Phase (degrees)
0
-1000
-2000
-3000
Frequency (kHz)

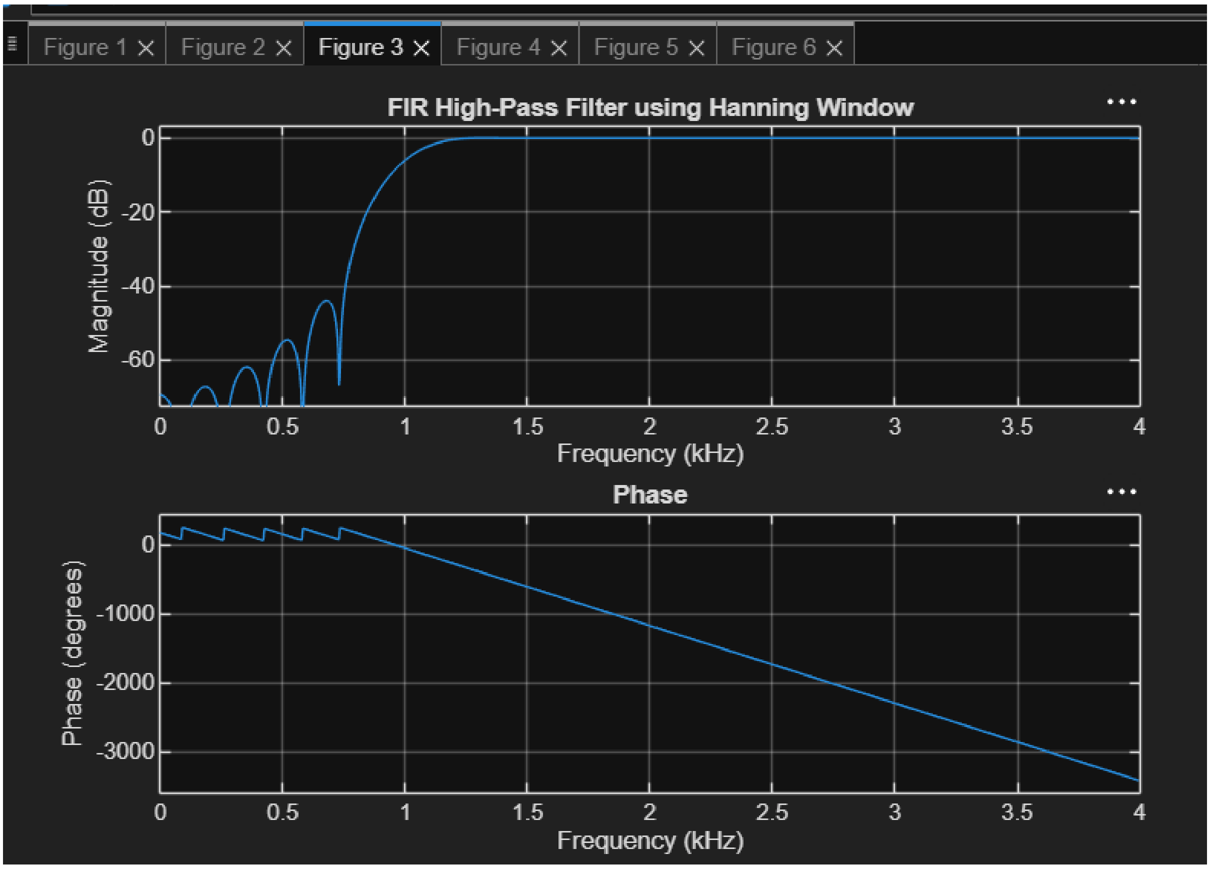
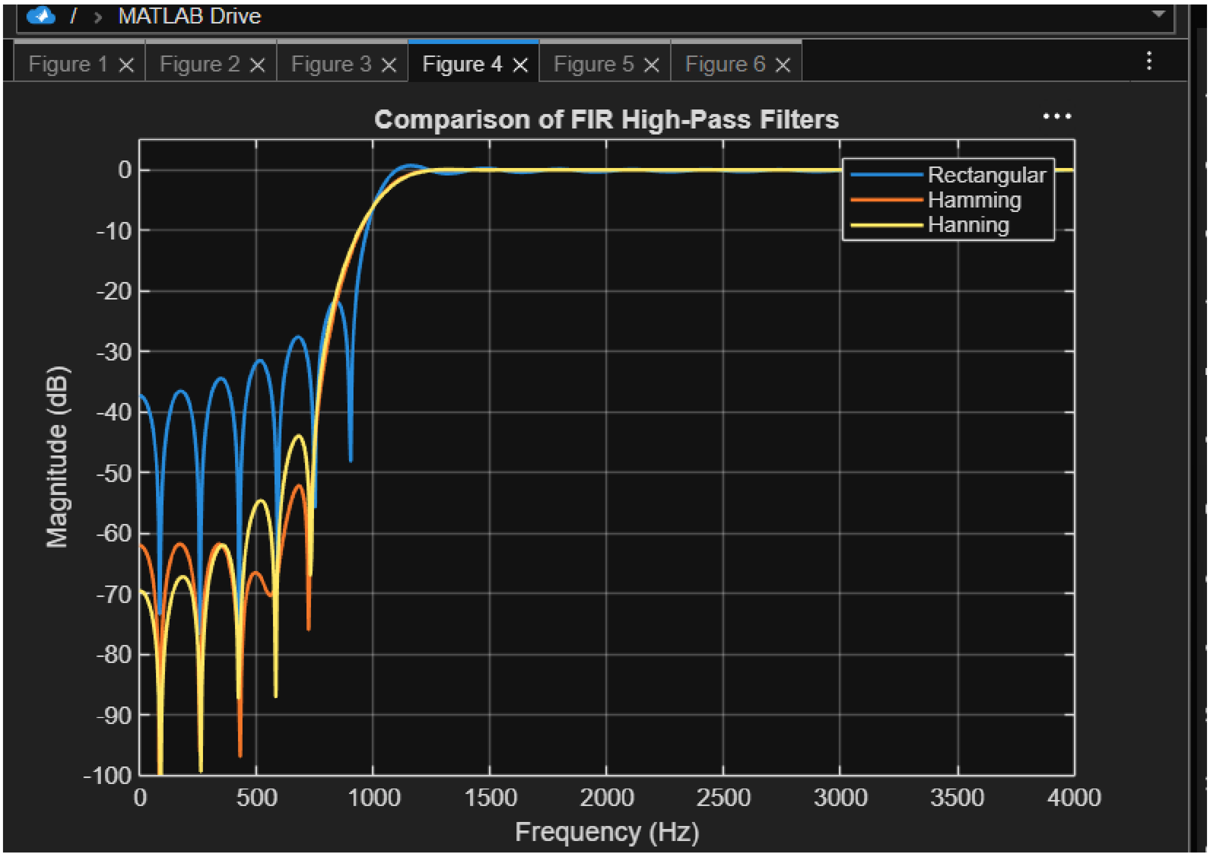


Figure 1 Figure 2 Figure 3 Figure 4 Figure 5 Figure 6
FIR High-Pass Filter using Hanning Window
Magnitude (dB)
0
-20
-40
-60
0 0.5 1 1.5 2 2.5 3 3.5 4
Frequency (kHz)
Phase
Phase (degrees)
0
-1000
-2000
-3000
0 0.5 1 1.5 2 2.5 3 3.5 4
Frequency (kHz)


/ › MATLAB Drive
Figure 1 Figure 2 Figure 3 Figure 4 Figure 5 Figure 6
Comparison of FIR High-Pass Filters
Rectangular
Hamming
Hanning
Magnitude (dB)
0
-10
-20
-30
-40
-50
-60
-70
-80
-90
-100
0 500 1000 1500 2000 2500 3000 3500 4000
Frequency (Hz)

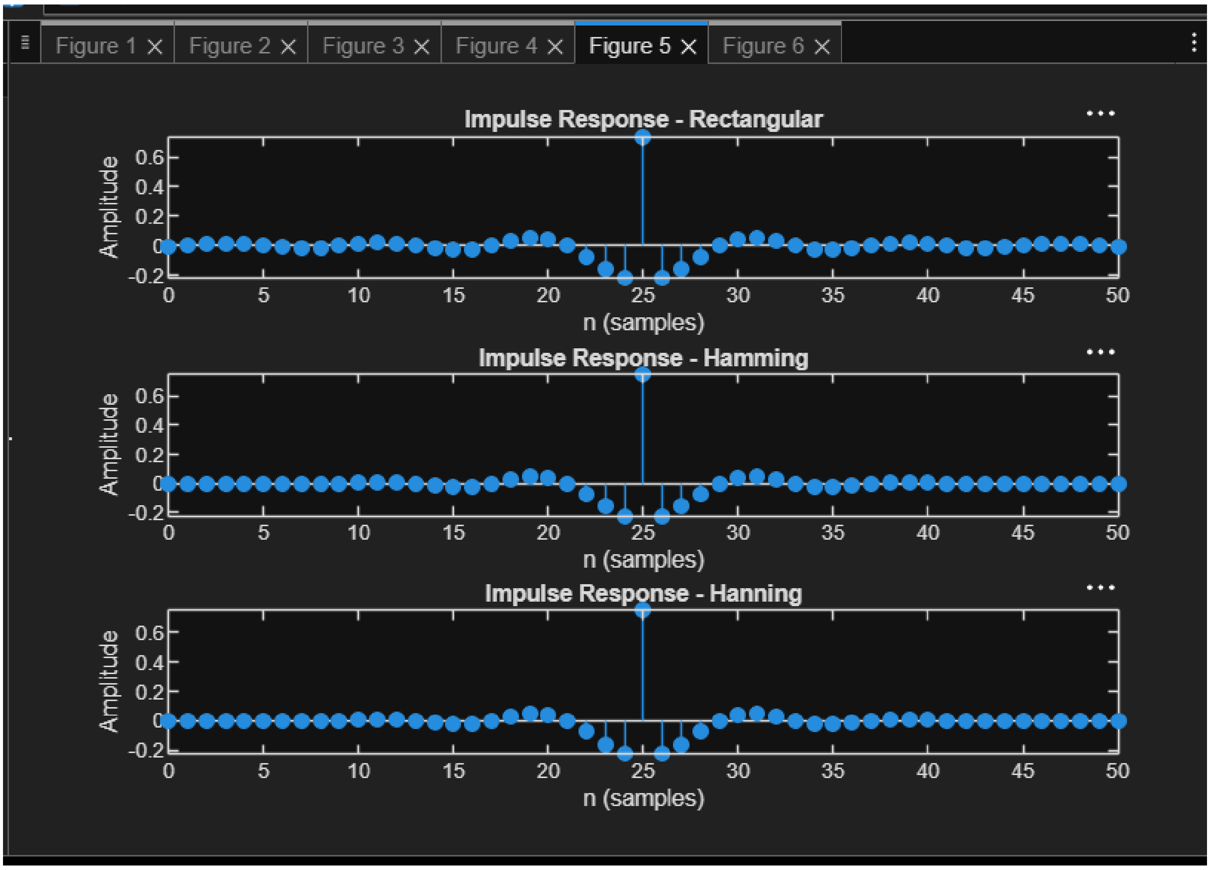


Figure 1
Figure 2
Figure 3
Figure 4
Figure 5
Figure 6
Impulse Response - Rectangular
Impulse Response - Hamming
Impulse Response - Hanning
Amplitude
n (samples)

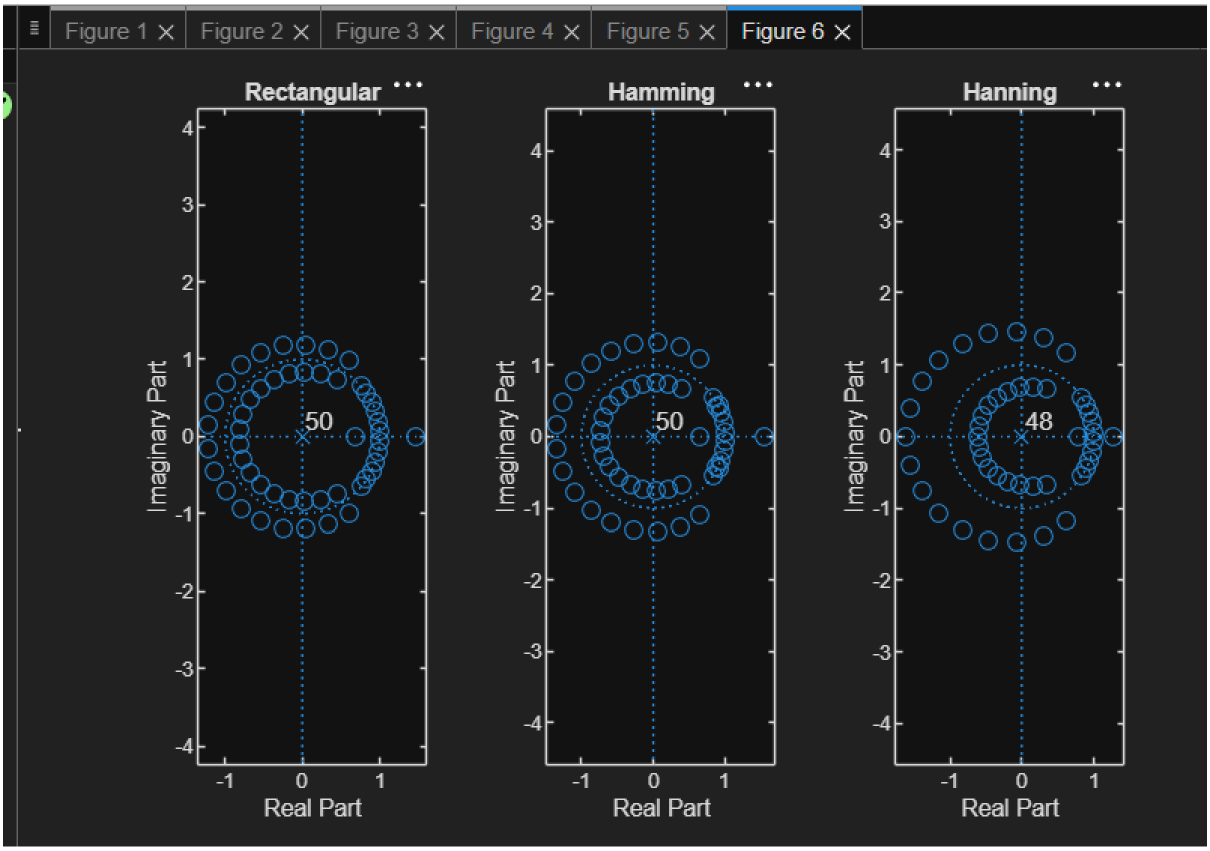


Figure 1
Figure 2
Figure 3
Figure 4
Figure 5
Figure 6
Rectangular
Hamming
Hanning
50
50
48
Imaginary Part
Real Part